Yeast9 Delta G Provenance

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✓ read and checked ■ read, provisional ✗ agent-written, not yet read

1 2026.09.10 - Where yeast-GEM 9's deltaG columns came from, and why they disagree with eQuilibrator ✗

1.1 The answer, first ✗

H. G. Martin asked why yeast-GEM 9.0.2's shipped `model_metDeltaG.csv` and `model_rxnDeltaG.csv` disagree with our eQuilibrator recomputation at all, since yeast9 almost certainly used a group-contribution method too. **The premise is right and the disagreement is still expected, for three separate reasons, and only the third is about thermodynamics.**

1. **The shipped columns are in kcal/mol and are labeled kJ/mol.** They are yETFL's values copied verbatim by `yeast-GEM/code/modelCuration/v8_7_0.m` (PR #330), and yETFL's model carries the field `thermo_units = 'kcal/mol'`. The yeast9 paper reports them as kJ/mol.
2. **The metabolite column was flattened across compartments.** matTFA transforms each formation energy at its own compartment's pH and ionic strength, so yETFL holds a different value per compartment. `v8_7_0.m` strips the compartment tag from the metabolite name, de-duplicates, and matches on the chemical, so every compartment copy gets one number. Exactly half the entries end up carrying another compartment's value.
3. **The estimator is a different group-contribution method.** yETFL used the group contribution method of Mavrovouniotis 1990 as packaged in matTFA's `thermo_data.thermodb`, with Alberty's transform, at 298.15 K, no magnesium. We use eQuilibrator 3 component contribution at 303.15 K, ionic strength 0.25 M, pMg 3.0. Different decomposition, different training corpus, different transform.

Does matching the conditions close the gap? For transport reactions yes, for chemistry no. With the units corrected and our table rebuilt at yETFL's own compartment conditions:

comparison	n	median abs diff (kJ/mol)	Pearson r
single compartment, as published	1,231	16.25	0.537
single compartment, units + conditions matched	1,230	**17.60**	0.513
transport (shipped nonzero), as published	752	17.36	0.298
transport (shipped nonzero), units + conditions matched	752	**5.61**	0.751
transport, all in comparison, matched	1,023	0.57	0.739

The horizontal arm of the cross in Figure 3 was never a disagreement about thermodynamics. It was two builds holding different compartment conditions plus a factor of 4.184. The single-compartment cloud is the real estimator difference and does not move.

Source: `experiments/026-metabolism-flux/results/thermo_yeast9_provenance.json`, written by `experiments/026-metab` `--report` after `--recompute` under the eQuilibrator environment.

1.2 The provenance chain, read out of the artifacts ✗

`v8_7_0.m` assigns, in preference order:

- `metDeltaG`: yETFL `metDeltaGFtr` matched by metabolite name with the compartment tag stripped; else ModelSEED `deltag` matched by `kegg.compound` id.
- `rxnDeltaG`: yETFL `rxnDeltaGR` matched by reaction id; else dGPredictor assigned by row position, `model.rxnDeltaG(3983:4062) = dgpred_rxnG.detaG(3983:4062)`.

yETFL is Oftadeh et al. 2021 (`10.5281/zenodo.4778047`), which states: "Using the group-contribution method (GCM), we estimated the Gibbs free energies of formation for 1092 of 1326 total unique metabolites." Its Methods cite Mavrovouniotis 1990 for the method and Alberty 1998 for the transform, and note that the standard Gibbs free energy of reactions with at least one metabolite in a membranous compartment, 1304 reactions, was not calculated by that procedure. dGPredictor is Wang et al. 2021, a Bayesian ridge fit (`M12_model_BR.pkl`) over radius-1 and radius-2 molecular signatures, driven at pH 7.0, ionic strength 0.1 M and 298.15 K, reporting kJ/mol.

Measured facts.

- Units: regressing our kJ/mol formation energies on yETFL's over 1,927 shared metabolites gives slope 4.187 through the origin at r = 0.987, against 4.184 kJ/kcal. yETFL prices cytosolic H2O at -37.21 and ATP at -564.41 against Alberty's -157.6 and -2292 kJ/mol.
- Identity: the shipped reaction column IS yETFL's `rxnDeltaGR` on 3,177 of 3,210 reactions, max abs difference 5.0e-3, which is what `round(x,2)` leaves. The 33 others are the dGPredictor block, in kJ/mol, so the column mixes two unit systems.
- Compartment collapse: 1,162 of 2,324 shipped metabolite values equal their own compartment's yETFL value. p75 error 86.7 kJ/mol, mean 74.5 kJ/mol.

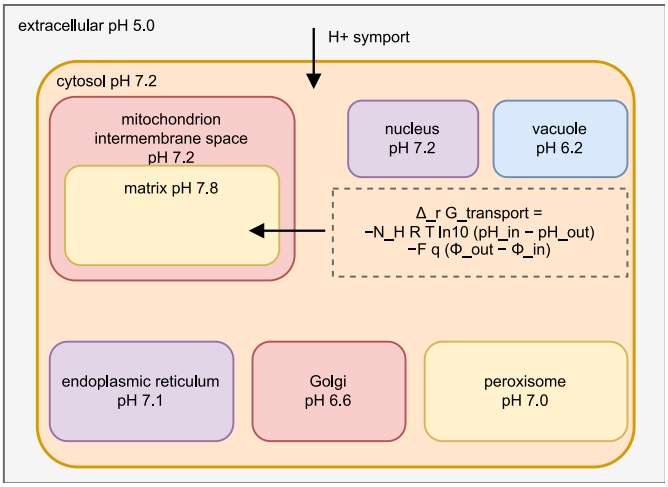
1.3 The "unexplained 10.2" is yeast-GEM disagreeing with itself ✗

`FluxLayer._transport_term` recorded a median 10.2 residual between the metabolite-sum energy and the shipped reaction column on single-compartment reactions as unexplained. It is fully explained, and it is in kcal/mol.

metabolite values summed	proton	single compartment	transport
shipped column	included	10.21 (n=2,021)	2.91 (n=1,183)
shipped column	dropped	2.37	2.78
yETFL per compartment	included	9.54 (n=2,005)	0.0039 (n=1,169)
yETFL per compartment	**dropped**	**0.0024**	**0.0038**

matTFA prices the proton at `-RT ln(10^-pH) = 9.54 kcal/mol` and then excludes it from every reaction sum (`if ~strcmp(metFormulas{i},'H')` in `prepModelforTFA.m`), because the transform already carries the reaction's pH dependence. yeast-GEM ships that number in the metabolite column anyway, so a consumer that sums the column pays it once per net proton. Within yETFL's own model the identity is exact: over 2,677 single-compartment reactions, `sum_i S_ij dfG'_i` with the proton dropped reproduces `rxnDeltaGR` to max abs residual 0.0.

1.4 Compartment conditions ✗



The compartments at the pH values this pipeline applies, with the two transport steps and the two terms of the transport free energy. Source `notes/assets/drawio/yeast-compartments-ph.drawio`; exported headless per `drawio-headless-export-gilahyper` and cropped with `gs -sDEVICE=bbox` then `pdfwrite`. This is panel a of Figure 4 in the 026 document.

Every pH in use is a textbook value. **The library holds no primary measurement of any yeast compartment pH.** The only in-library sources are two modeling papers that tabulate them as parameters: `smithThermofluxGenerationAnalysis2026` Table 1 (cytosol 7.0, mitochondrion 7.4, extracellular 5.0, pMg 14, T 298 K, I 0.25 M, plasma membrane 60 mV, mitochondrial membrane 160 mV) and `oftadehGenomescaleMetabolicModel2021` through its `yeast_compartment_data.json`. Nothing in the 026 document cites either through bibtex, because that document's `references.bib` is deliberately empty and works are named in prose.

comp	name	pipeline pH / I / Phi	yETFL pH / I / Phi	literature in library
c	cytoplasm	7.2 / 0.25 / 0	7.00 / 0.25 / 0	no primary source
ce	cell envelope	5.0 / 0.25 / +60	7.00 / 0 / 0	no primary source
e	extracellular	5.0 / 0.25 / +60	5.00 / 0 / +60	no primary source
er	endoplasmic reticulum	7.1 / 0.25 / 0	7.20 / 0 / 0	no primary source
erm	ER membrane	7.1 / 0.25 / 0	7.20 / 0 / 0	no primary source
g	Golgi	6.6 / 0.25 / 0	6.35 / 0 / 0	no primary source
gm	Golgi membrane	6.6 / 0.25 / 0	6.35 / 0 / 0	no primary source
im	mito intermembrane	absent from yeast-GEM 9	7.00 / 0 / 0	no primary source
lp	lipid particle	7.2 / 0.25 / 0	7.00 / 0.25 / 0	no primary source
m	mitochondrion	7.8 / 0.25 / -160	7.50 / 0.25 / -180	no primary source
mm	mitochondrial membrane	7.8 / 0.25 / -160	7.50 / 0.25 / -180	no primary source
n	nucleus	7.2 / 0.25 / 0	7.00 / 0 / -15	no primary source
p	peroxisome	7.0 / 0.25 / 0	8.20 / 0 / 0	no primary source
v	vacuole	6.2 / 0.25 / 0	6.17 / 0 / 0	no primary source
vm	vacuolar membrane	6.2 / 0.25 / 0	6.17 / 0 / 0	no primary source

pH is dimensionless, I in M, Phi in mV signed against the cytosol. Thermo-Flux Table 1 gives potential magnitudes rather than signs; the signs in the pipeline column are assigned from the direction of the gradients they describe. Written by `yeast9_delta_g_provenance.py --compartments` to `results/compartment_conditions.json`.

Curation requests (identified, not added; adding needs an explicit go-ahead):

- Orij et al. 2009, in vivo cytosolic and mitochondrial pH in *S. cerevisiae* by ratiometric pHluorin.
- Brett et al. 2005 and Plant et al. 1999, vacuolar lumen pH in *S. cerevisiae*.
- Llopis et al. 1998, organelle pH by targeted fluorescent protein (matrix, Golgi, ER).
- A peroxisomal pH measurement for *S. cerevisiae*. The two tables in use differ by 1.2 pH units (7.0 vs 8.2) and nothing in the library settles it.

1.5 The transport term, computed ✗

`torchcell/metabolism/transport.py` computes Thermo-Flux Eq. (7) per reaction from the stoichiometry, the compartment of each metabolite and its charge.

Sign convention, fixed here because the paper does not fix it. A species moves from the compartment it appears in as a reactant (`out`) to the one it appears in as a product (`in`), as written. `DeltapH = pH_in - pH_out` and `DeltaPhi = Phi_out - Phi_in`, so

```
Delta_r G^t = -N_H R T ln(10) (pH_in - pH_out) + q F (Phi_in - Phi_out)
```

which is the paper's `-N_H RT ln10 DeltapH - F q DeltaPhi` with those two definitions. This matches matTFA's `calcDGtpt_RHS`, where the tabulated pair potential is `Phi_in - Phi_out` (yETFL's `membranePot['c']['m'] = -180`, matrix negative relative to cytosol). Choosing the other sign inverts every proton pump with no other symptom.

`N_H` counts only the explicit H+ species yeast-GEM carries per compartment. Thermo-Flux's `N_H` also absorbs the protons released or bound when a reactant's average protonation differs across a pH gradient; that needs a species distribution per compound and is not computed here.

Results on yeast-GEM 9.0.2 (`results/transport_terms.json`, `results/transport_terms_yeast_gem_9.json`):

- 1,468 reactions cross a compartment boundary. 326 get a nonzero term, 183 move a proton, 482 move charge.
- Against the shipped reaction column converted to kJ/mol, on the 287 reactions nonzero in both: sign agreement 0.847, Pearson r 0.804, median abs difference 7.04 kJ/mol against a median magnitude of 12.0, p90 18.9, max 120.8.
- Under yETFL's own compartment conditions instead: 299 in both, sign 0.823, r 0.813, median abs difference 6.08 kJ/mol. So the residual is mostly method, not conditions.

Where they disagree. 613 reactions have a nonzero shipped value and a zero computed one. Those are the protonation term: a reactant with no explicit proton and no charge still changes its transformed formation energy crossing into a different pH, and that is the contribution left out above. 11 go the other way, computed nonzero against a shipped zero, and those are reactions yETFL declined to price rather than priced as zero.

Wiring. `FluxLayerConfig.transport_delta_g_source` takes `off` (the default, and what every 026/027 arm ran), `shipped` (recover from the GEM's own reaction column on the reactions whose metabolite-sum energy is structurally zero; must now be read as kcal), and `computed` (read the table `transport.py` writes, at `transport_terms_path`). Only `computed` carries to a model that ships no thermodynamics. `use_shipped_transport_delta_g` is kept as a deprecated alias so existing arm configs still mean what they meant.

1.6 Why there is an "ours" at all instead of calling eQuilibrator per reaction ✗

The transform is not reimplemented. `Compound.transform` and `ComponentContribution.standard_dg_formation` are eQuilibrator's own. What is ours is the assembly: the stoichiometric sum, the alignment to the GEM's metabolite index, and the zero-proton convention.

The assembly exists because the flux layer consumes metabolite-level quantities. `FluxLayer.reaction_potential` in ANCHORED mode forms `delta_r_g0 + RT * S^T log c + delta_r_g_transport + delta_g_offset` with `log c` a learned per-metabolite vector, and in FREE mode `S^T mu` with `mu` per metabolite. A reaction-level `Delta_r G'^o` cannot be decomposed back into either, and the square-root covariance factor Sigma that gives `Q` is per metabolite as well. So the per-metabolite table is justified, and panel c of Figure 4 should be read as a check of the assembly rather than as a verification of a reimplemented transform. The caption now says that.

1.7 Mirrored upstream artifacts ✗

Copied to `$DATA_ROOT/data/torchcell/yetfl_thermo/` with sha256 recorded in `results/thermo_yeast9_provenance.json` under `upstream_artifacts`, each with its `source_url` and `retrieval_command`:

file	sha256	source
`yeast8_thermo_curated.mat`	`ec82ce09e76e5b5fc8a75fc86f86eb3c8510b1f5d[illegible]`	`[illegible]dels.zip`
`yeast_compartment_data.json`	`3b96940ad301622309758d501ac1276cfa90e1c60[illegible]`	`[illegible]ast.zip`
`dgpred_rxnG.csv`	`0870ec4d64773fcea0fd6249d7922a42d4bc0af19[illegible]`	`[illegible]uration/v8_7_0/`
`v8_7_0.m`	`7c8bfb5b0fd5f53e89ea182f528a30b18dd5938a[illegible]`	`[illegible]uration/`
`run_dgpredictor.py`	`1d60f1c9963dd954ea7b888384f146b155e777a0[illegible]`	`data/modelCuration/v8_7_0/code/`

1.8 Files ✗

- `torchcell/metabolism/transport.py`
- `tests/torchcell/metabolism/test_transport.py`
- `torchcell/metabolism/flux_layer.py` (`_transport_term`, `transport_delta_g_source`, `transport_terms_path`)
- `experiments/026-metabolism-flux/scripts/yeast9_delta_g_provenance.py`
- `experiments/026-metabolism-flux/scripts/plot_equilibrator_thermo.py`
- `experiments/026-metabolism-flux/results/thermo_yeast9_provenance.json`
- `experiments/026-metabolism-flux/results/transport_terms.json`
- `experiments/026-metabolism-flux/results/transport_terms_yeast_gem_9.json`
- `experiments/026-metabolism-flux/results/compartment_conditions.json`
- `notes/assets/drawio/yeast-compartments-ph.drawio`
- `notes-tex/026-kinetics-thermodynamics/sections/3-thermodynamics.tex`

1.9 Open ✗

- The protonation contribution to `N_H` needs an eQuilibrator species distribution per transported compound. That is the 613-reaction shortfall above.
- Whether to correct the shipped column's unit at read time in `torchcell/metabolism/constraints.load_thermo_table`, which currently records `units = "kJ/mol"` for a kcal/mol file. Not changed here: that file belongs to another work item, and changing it silently rescales every arm that used `shipped`.
- Report the unit error upstream to SysBioChalmers/yeast-GEM.